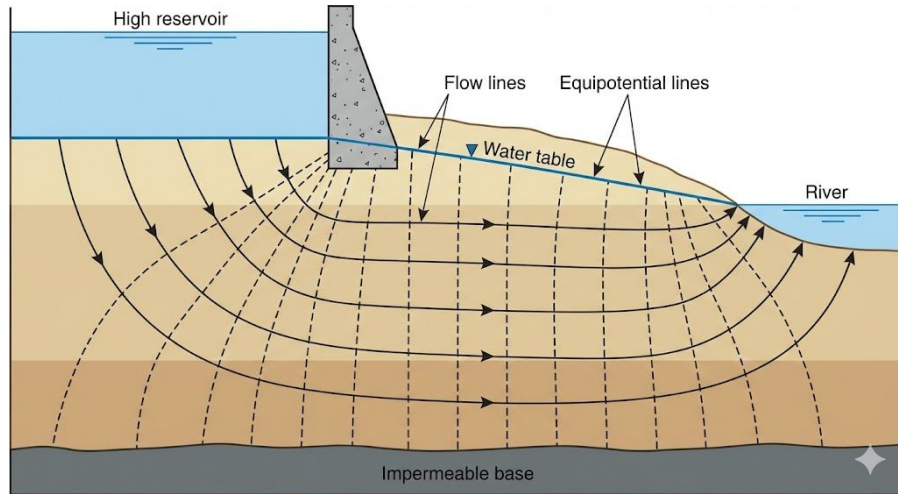




Chapter-3 Potential Groundwater Flow Theory and Flow Net Analysis



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- Groundwater Flow Visualisation
- Groundwater Flow Direction & Mapping
- Governing Equation of Groundwater Flow
- Steady Flow in Unconfined Aquifer
- Unconfined Aquifer with Recharge
- Steady Flow in Confined Aquifer
- References

Need for Visualisation: Understanding subsurface flow is complex due to the invisibility of the medium. Visualisation bridges the gap between mathematical abstraction and physical reality.

Flow Paths & Energy: Visualization helps delineate how water moves from recharge to discharge areas and how energy (hydraulic head) dissipates through the system.

(Freeze & Cherry, 1979)

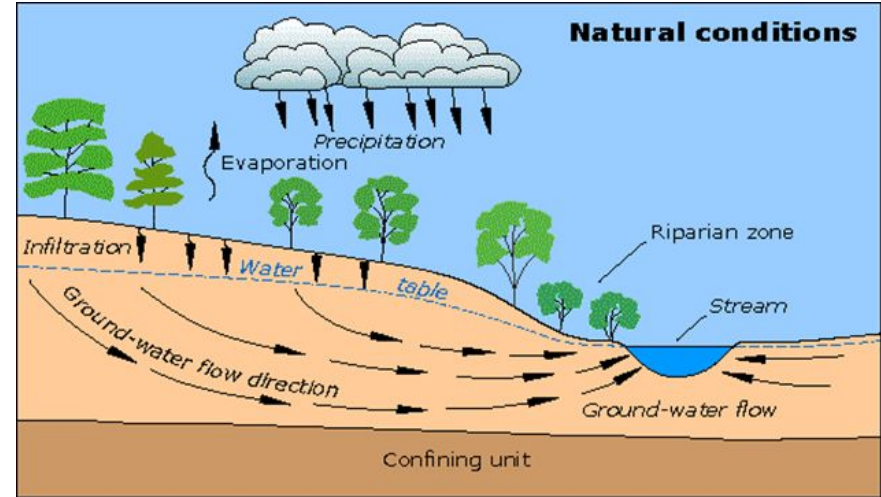


Figure: Groundwater flow showing natural conditions, Water Science School, USGS

- Definition: An imaginary line that traces the path a particle of water follows as it flows through an aquifer (Fetter, 2001).
- Direction: Always tangent to the velocity vector of the flow at every point.
- Properties: Streamlines cannot cross each other in steady flow (laminar flow assumption).

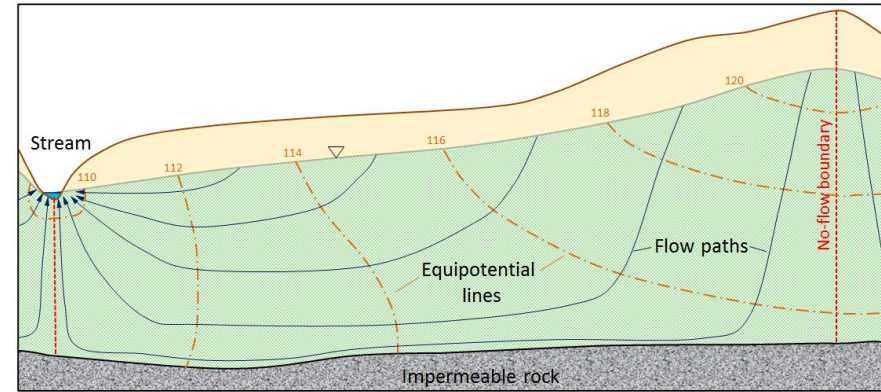


Fig: Predicted equipotential lines (orange) and groundwater flow paths (blue) in an unconfined aquifer. Source: BCcampus Open Publishing, 2015

Plotting & Mathematical Visualisation

To plot **streamlines**, we use the Stream Function (Ψ). In a 2D isotropic aquifer, streamlines are the curves where Ψ is constant.

Orthogonality: In isotropic media, streamlines are always perpendicular to equipotential lines (lines of constant hydraulic head, h).

Flow Nets: When you plot streamlines and equipotential lines together, they form a "flow net."

Spacing: The distance between two streamlines represents a constant discharge (Q). Where streamlines converge (get closer), the flow velocity is increasing.

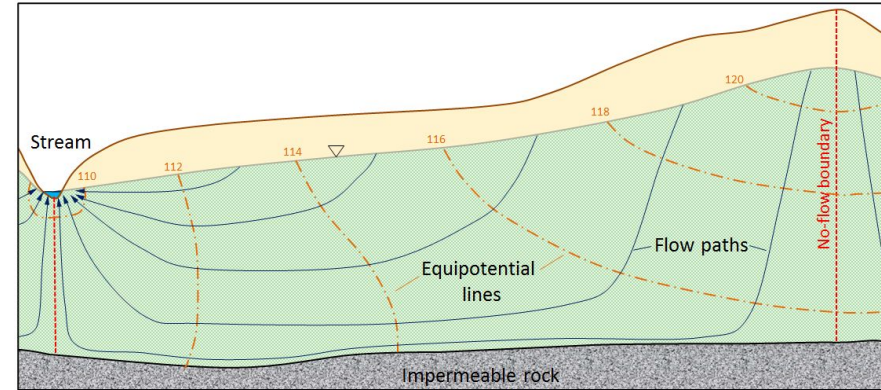


Fig: Predicted equipotential lines (orange) and groundwater flow paths (blue) in an unconfined aquifer. Source: BCcampus Open Publishing, 2015



- **Definition:** Lines (or surfaces in 3D) connecting points of equal total hydraulic head (h) (Todd & Mays, 2005).
- **Hydraulic Head:** Represents the mechanical energy per unit weight of the fluid.
- **Properties:** Work done in moving a unit mass of water along an equipotential line is zero.

A line along which velocity potential function (ϕ) is constant is known as the equipotential line.

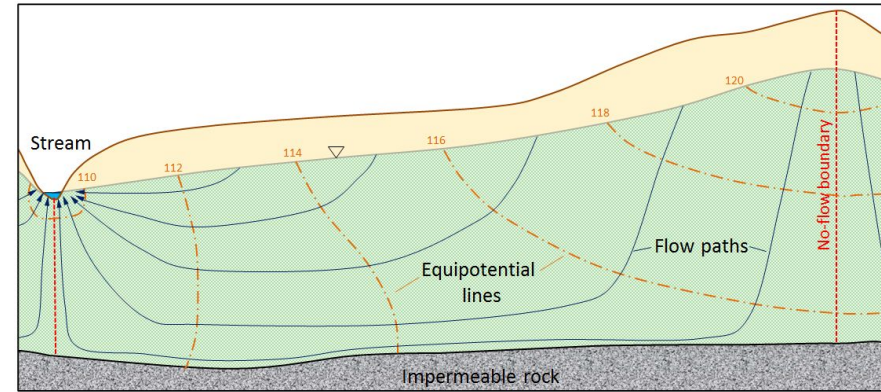


Fig: Predicted equipotential lines (orange) and groundwater flow paths (blue) in an unconfined aquifer. Source: BCcampus Open Publishing, 2015

Equipotential Lines (Plotting)

- Calculated by evaluating total hydraulic head $h = z + p/\gamma$ at discrete points.
- Points with equal head values are connected to form equipotential lines (2D) or surfaces (3D).
- Typically plotted as head contours using field measurements or numerical models.
- Equipotential lines intersect flow lines at right angles in isotropic media.

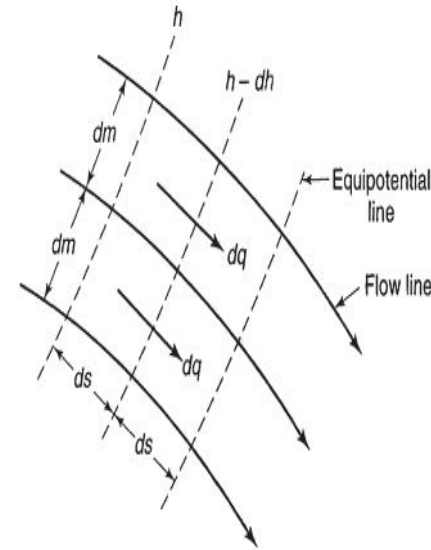


Figure 3.6.1. Portion of an orthogonal flow net formed by flow and equipotential lines.

Fig: Equipotential lines and groundwater flow paths.
Source: Todd and Mays, 2005



- **Orthogonality:** In an isotropic medium, streamlines and equipotential lines intersect at right angles (90°) (Bear, 1972).
- **Flow Direction:** Groundwater always flows from higher hydraulic head to lower hydraulic head, perpendicular to the equipotential lines.
- **Physical Significance:** This relationship is the basis for graphical solutions to flow problems.

$$\left(\frac{dy}{dx}\right)_\phi \times \left(\frac{dy}{dx}\right)_\psi = -1$$

Slope of equipotential Line $\times$ Slope of stream function = -1

For a streamline, $\psi(x, y) = \text{constant}$ and the differential of ψ is zero.

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$$

$$d\psi = -v dx + u dy$$

$$\left(\frac{\partial y}{\partial x}\right)_{\psi=\text{const}} = \frac{v}{u}$$

For an equipotential line, $\phi(x, y) = \text{constant}$ and the differential of ϕ is zero.

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy$$

$$d\phi = u dx + v dy$$

$$\left(\frac{\partial y}{\partial x}\right)_{\phi=\text{const}} = -\frac{u}{v}$$

$$\left(\frac{\partial y}{\partial x}\right)_{\psi=\text{const}} = -\frac{1}{\left(\frac{\partial y}{\partial x}\right)_{\phi=\text{const}}}$$

Definition: A graphical representation of a flow field comprising a family of streamlines and a family of equipotential lines (Cedergren, 1989).

Components:

- **Flow Channels:** Space between two adjacent streamlines.
- **Potential Drops:** Difference in head between two adjacent equipotential lines.
- **Conditions:** Must satisfy Laplace's equation and boundary conditions.

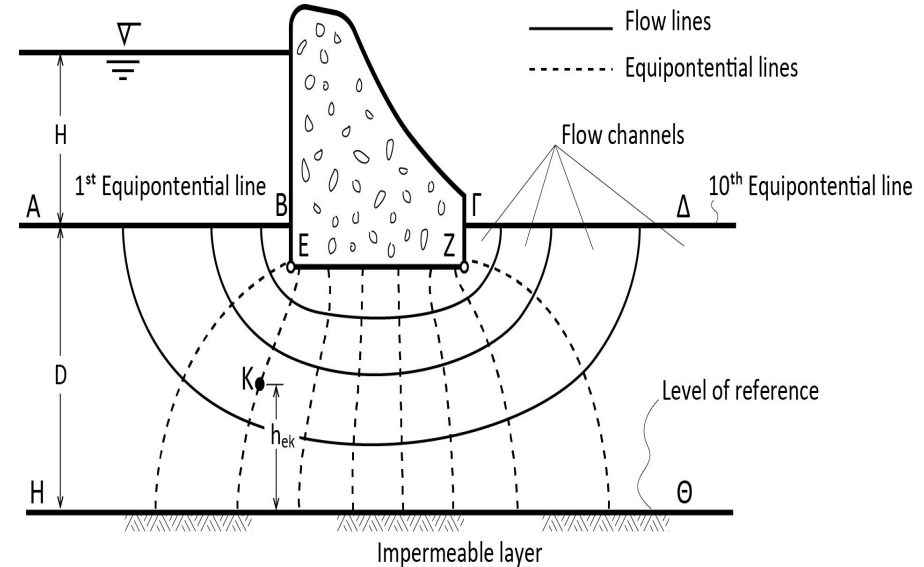


Fig: Graphical generation of flow nets; Source: Athanasopoulos, G. A. (2021) "Online Lecture Notes on Soil Mechanics"



Boundary Conditions: Identify impervious boundaries (streamlines) and constant head boundaries (equipotential lines).

Method of Plotting: Trial-and-error sketching to form "curvilinear squares."

Curvilinear Squares: The shapes formed by the intersection should approximate squares (mean width is almost equal to mean length) (Das, 2013).

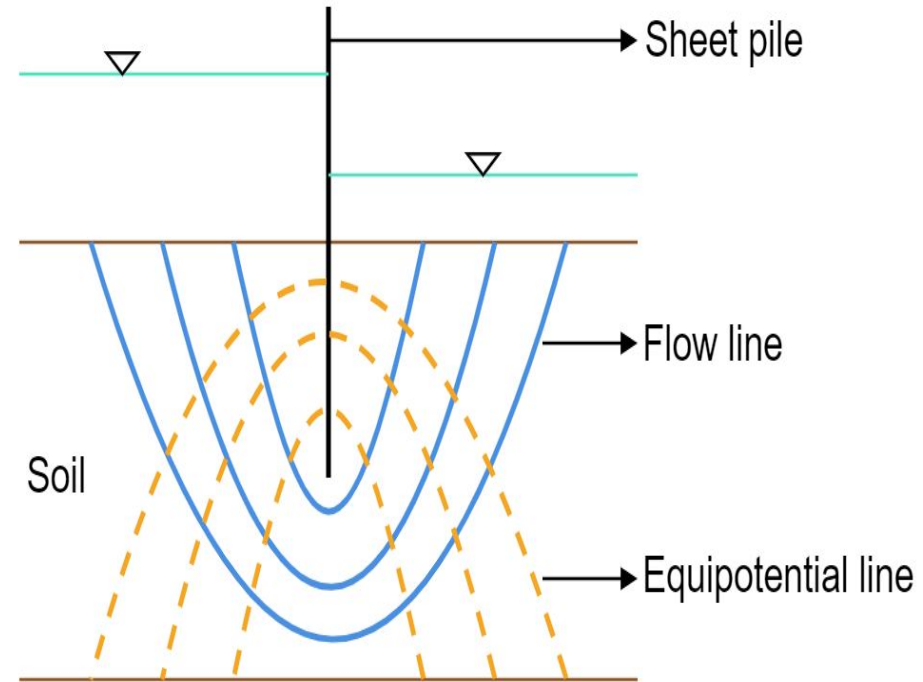


Fig: Flownet Characteristics, Source: APSEd, n.d.

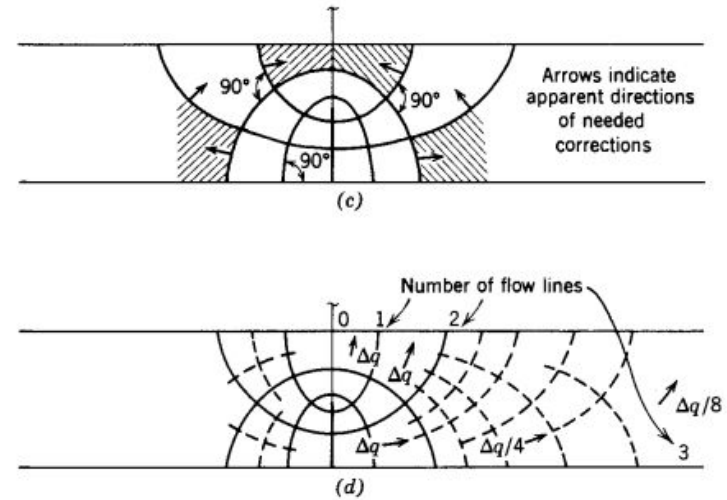
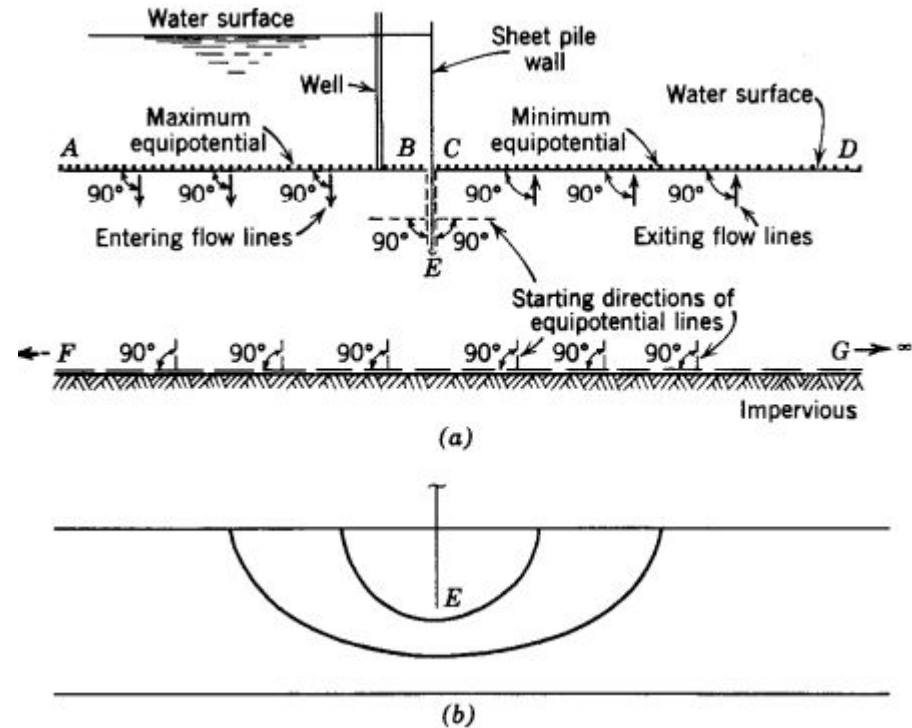


FIG. 4.2 Example of the construction of a flow net for confined flow in single permeability section (Example 1). (a) Identify prefixed conditions, noting starting directions of lines. (b) Draw trial family of flow lines (or equipotentials) consistent with prefixed conditions. (c) Keeping the lines drawn in (b), sketch first trial flow net. Make all lines intersect other set of lines at 90°. (d) Erase and redraw lines until all figures are square. Subdivided as desired for detail and accuracy.

Source: Cedergen, 1989



Seepage Estimation: Calculating the quantity of seepage (Q) under dams or through levees. $Q = k H (N_f / N_d)$

Hydraulic Gradient: Determining the exit gradient to assess piping risk.

Head at any point: Flow net can be used to determine the head at any point. H/N_d gives the head drop between two consecutive equipotential lines.

Engineering Applications: Stability analysis of hydraulic structures and uplift pressure calculation (Todd & Mays, 2005).



Definition: The sum of elevation head (z) and pressure head (P/γ) at a specific point (Freeze & Cherry, 1979).

$$h = z + P/\gamma$$

Measurement: Measured using piezometers (observation wells) penetrating the aquifer.

Importance: It is the driving potential for groundwater movement.

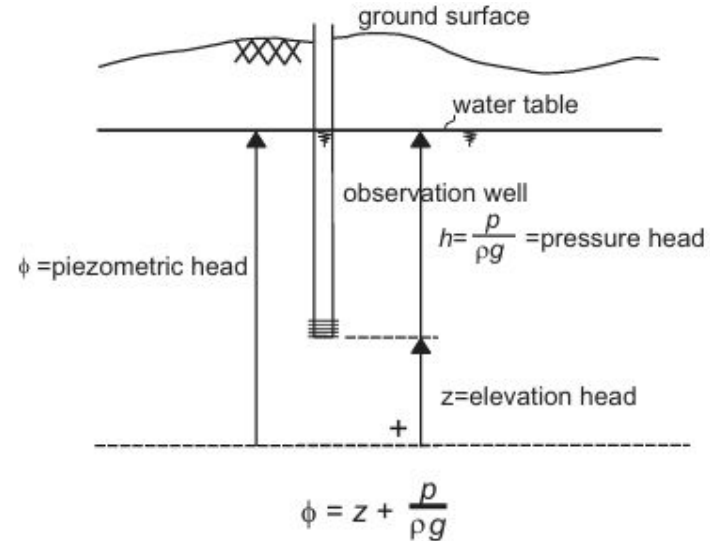


Figure: Piezometric Head, Source: Essink, 2001



Fundamental Law: Fluid flows from regions of higher total head to regions of lower total head.

Triangulation: Using three observation wells to determine the plane of the water table and flow direction (Heath, 1983).

Determination: The gradient of the head determines the exact vector of flow.

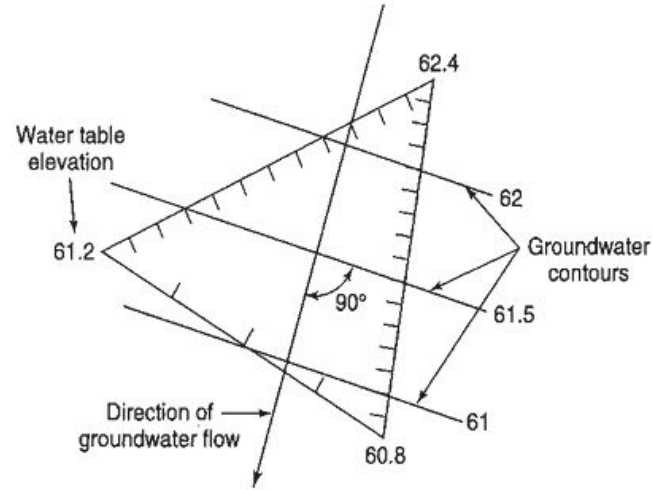


Figure 3.6.4. Estimate of groundwater contours and flow direction from water table elevations in three wells.

Source: Todd and Mays, 2005



Definition: The change in hydraulic head (dh) per unit distance (dl) in the direction of maximum decrease in head.

$$i = dh/dl$$

Relation to Velocity: Directly proportional to Darcy velocity ($v = -Ki$).

Significance: Steeper gradients imply faster flow velocities (Fetter, 2001)

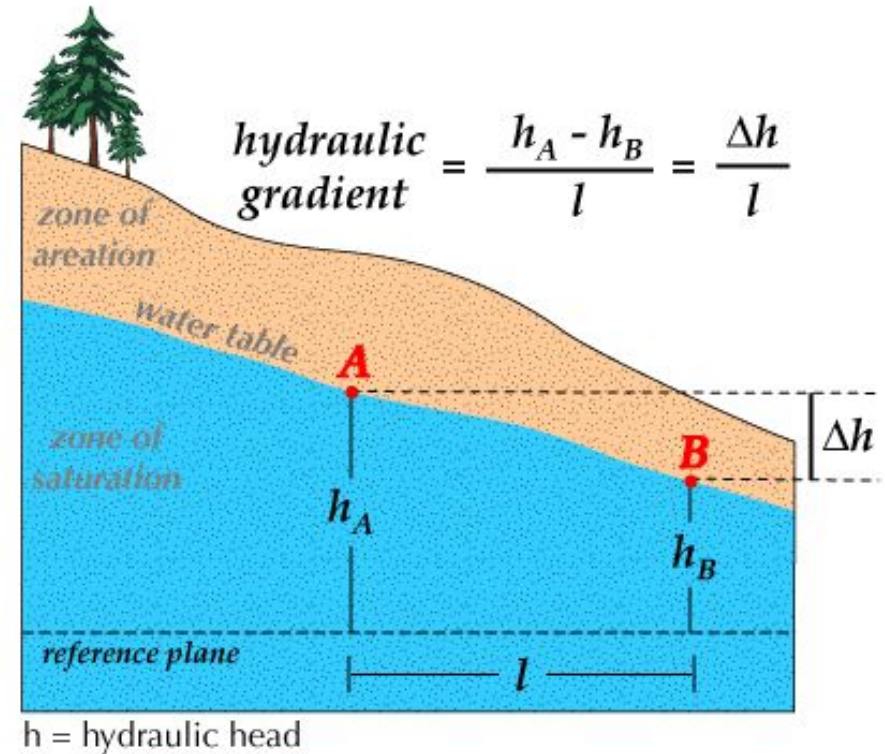


Figure: Hydraulic Gradient; Source: Punchlist Zero n.d.



Definition: The surface where pore water pressure equals atmospheric pressure (Freeze & Cherry, 1979).

Unconfined Aquifers: It marks the upper boundary of the saturated zone.

Factors Affecting: Topography, recharge rates, extraction (pumping), and seasonal variations.

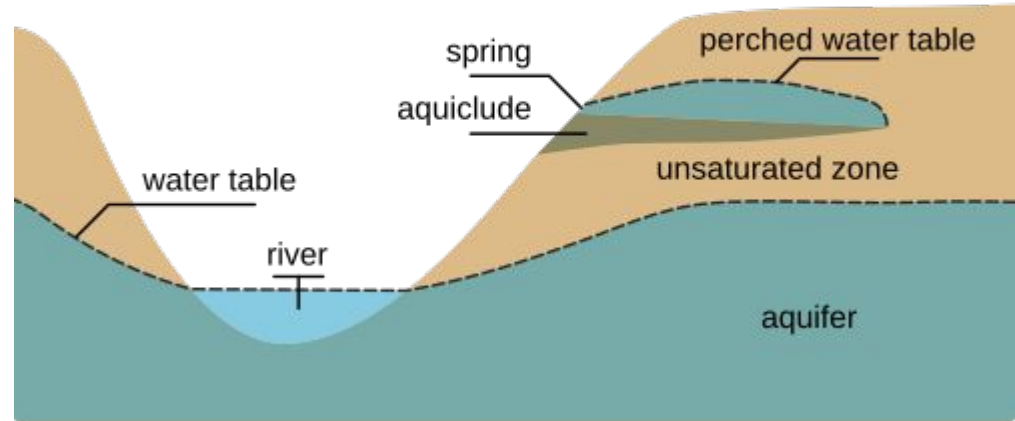


Figure: Water Table; Source: Wikipedia

Groundwater Flow Direction and Mapping: Water Table Maps



- **Preparation:** Plotting head data from multiple wells on a map.
- **Contour Mapping:** Drawing lines connecting points of equal head (equipotentials).
- **Data Requirements:** Precise elevation data of well casings and depth-to-water measurements (USGS standards).

Groundwater elevations from wells → **water level contour maps**

Flow direction is perpendicular to contour lines

Useful for understanding **groundwater movement patterns**

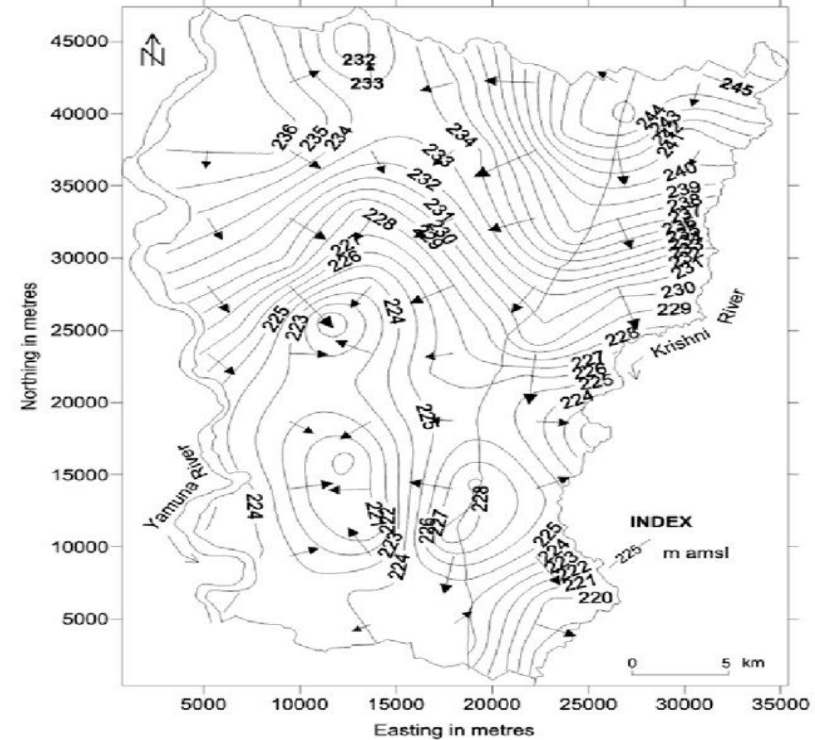
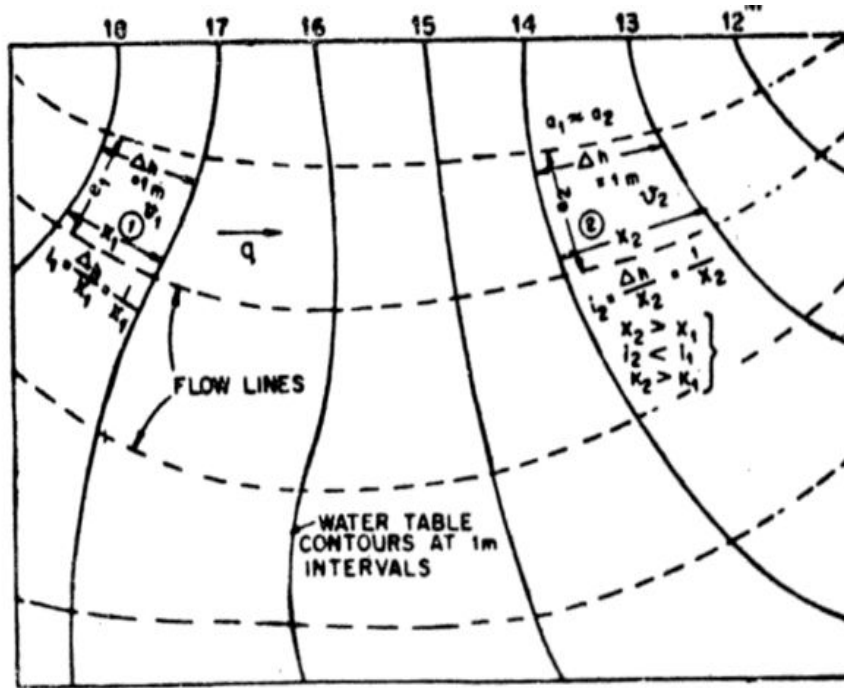


Figure: Water Table Map; Source: Ahmed & Umar, 2009

Groundwater Flow Direction and Mapping: Water Table Maps



SECTION 2 IS A MORE PROMISING SITE FOR A WELL

Fig. 4.8 Water table (piezometric surface) contour maps.

Since there is no flow across the flow lines

$$q = a_1 v_1 = a_2 v_2 = a_1 \cdot K_1 i_1 = a_2 \cdot K_2 i_2 \dots \text{from Darcy's law}$$

Therefore
$$\frac{K_1}{K_2} = \frac{a_2 i_2}{a_1 i_1}$$

$$\frac{K_1}{K_2} = \frac{i_2}{i_1}$$

when flow lines are nearly parallel.

If the spacing between the water table contours (contour interval Δh) at section 1 and 2 are x_1 and x_2 , respectively, then

$$i_1 = \frac{\Delta h}{x_1}, i_2 = \frac{\Delta h}{x_2}$$

when

$$\frac{K_1}{K_2} = \frac{x_2}{x_1}$$

which indicates that for areas with uniform ground water flow, the portions having wide water table contour spacing having flat gradients and higher permeabilities. Similarly a_2/a_1 can be estimated from the distances between the flow lines at the two sections. Thus section 2 indicates a better and more promising site for a well than section 1.

Such water table contour maps with flow lines drawn provide useful data for locating new wells as the best possible sources of ground water supply and areas of most favourable permeability can be interpreted from them.

Raghunath, H. M. (1987). Groundwater. Wiley Eastern.



Analysis of Watertable Maps

If the flow rates (q_1 and q_2) across successive watertable or piezometric surface contours between two limiting flow lines are known, Fig. 4.10 the following deductions can be obtained:

$$(i) \quad q_1 = T_1 i_1 w_1, \quad i_1 = \frac{\Delta h}{l_1}, \quad l_1 = MN, \quad w_1 = AA' \text{ (scaled from map)}$$

$$q_2 = T_2 i_2 w_2, \quad i_2 = \frac{\Delta h}{l_2}, \quad l_2 = PQ, \quad w_2 = BB' \text{ (scaled from map)}$$

Δh = water level difference between two successive contours

T_1 and T_2 can be estimated. Repeating this procedure spatial distribution of T can be obtained.

(ii) If there is no leakage from above or below into the aquifer the storage coefficient S can be estimated from

$$q_2 - q_1 = S \Delta h_t A_L \quad (4.19)$$

where Δh_t = average rate of water level decline or rise, in the area between the two limiting flowlines and successive contours, A_L .

(iii) Recharge by vertical leakage of water through an aquitard can be obtained from

$$PA_L = (q_2 - q_1) \pm S \Delta h_t A_L \quad (4.20)$$

where P = rate of vertical leakage per unit area;

+ sign is used for rise in water level and

− sign is used for decline in water level.

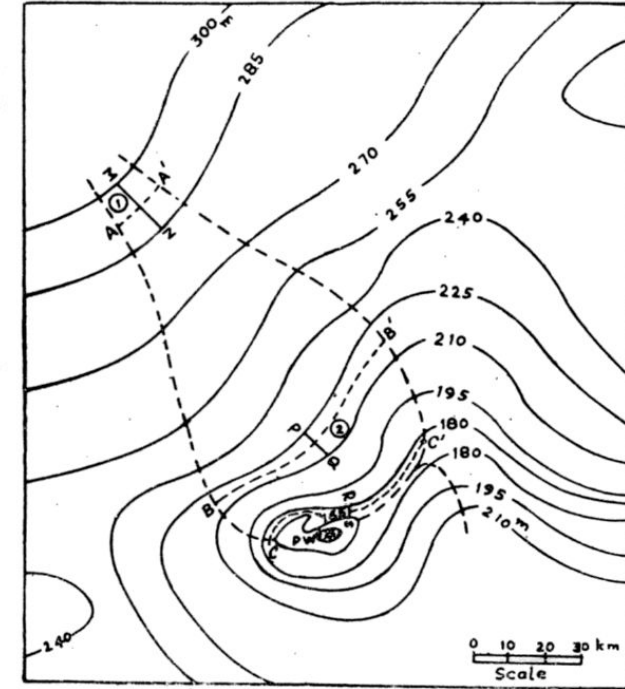
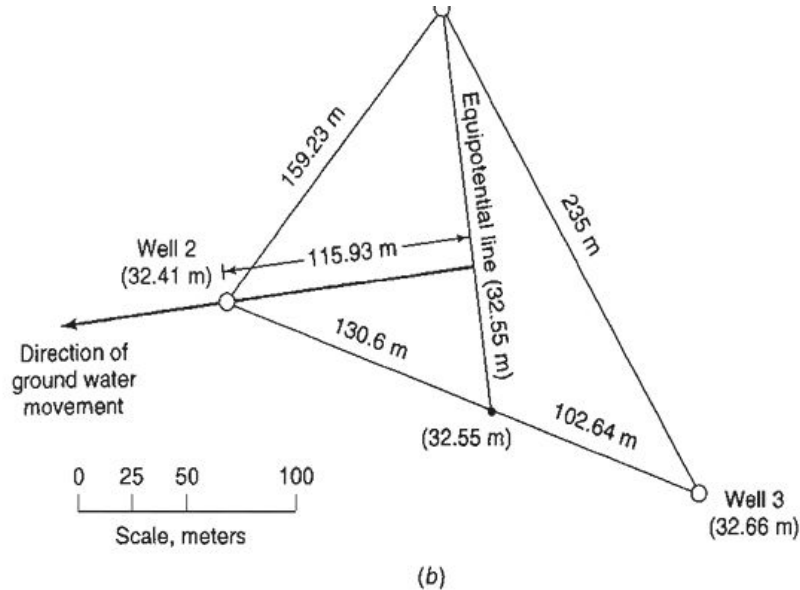


Fig. 4.10 Flow analysis from the contour map of GWT or piezometric surface.



EXAMPLE 3.6.1

Three observation wells are installed to determine the direction of groundwater movement and the hydraulic gradient in a regional aquifer. The distance between the wells and the total head at each well are shown in Figure 3.6.7a.



Step 1: Identify the well with the intermediate water level—Well 1 in this case.

Step 2: Along the straight line between the wells with the highest head and the lowest head, identify the location of the same head of the well from Step 1. Note that this is accomplished by locating the elevation of 32.55 m between Well 2 and Well 3 in the graphical solution.

Step 3: Draw a straight line between the intermediate well from Step 1 and the point identified in Step 2. This is a segment of the equipotential line along which the total head is the same as that in the intermediate well (i.e., equipotential line of 32.55 m head in this case).

Step 4: Draw a line perpendicular to the equipotential line passing through the well with the lowest head. The hydraulic gradient is the slope of that perpendicular line. Also, the direction of the line indicates the direction of groundwater movement. The graphical procedure above is illustrated in Figure 3.6.7b. The hydraulic gradient is then computed as

$$i = \frac{32.55 \text{ m} - 32.41 \text{ m}}{115.93 \text{ m}} = 0.0012$$

Source: Todd and Mays, 2005

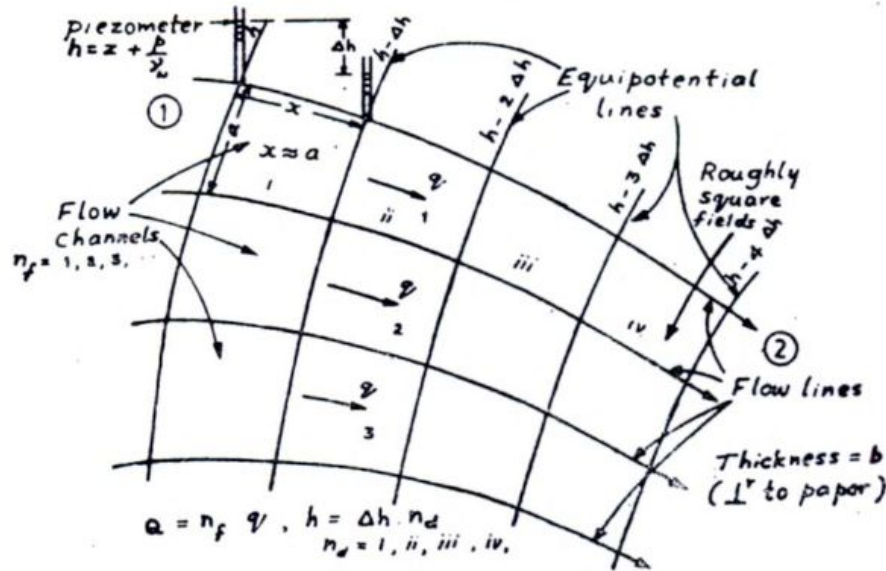


Fig. 4.11 Flownet in an isotropic aquifer.

Total head $h = h_1 - h_2$

Head drop between successive equipotential lines $\Delta h = \frac{h}{n_d} \dots (i)$

Flow through one flow channel by Darcy's law

$$q = K \cdot \frac{\Delta h}{x} \cdot (a \cdot b), \quad a \approx x \text{ for roughly square fields}$$

$$= K \Delta h \cdot b$$

Total flow through n_f flow channels

$$Q = q \cdot n_f = n_f (K \Delta h b), \quad T = K b \dots (ii)$$

$$Q = n_f \Delta h T$$

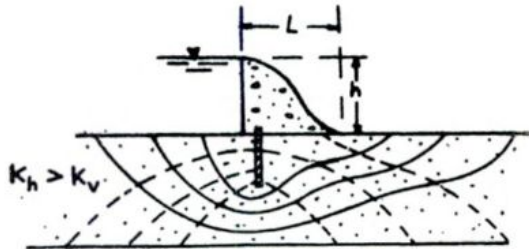
From (i) and (ii),

$$Q = \frac{n_f}{n_d} K h b$$

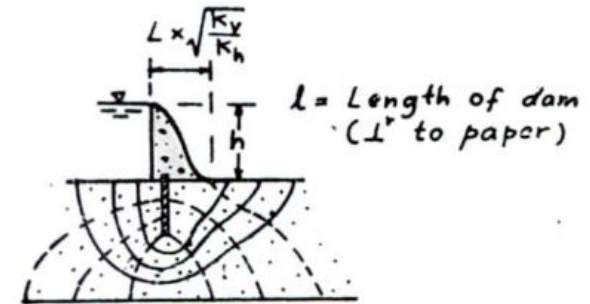


Flownet—Anisotropic Aquifer

For an anisotropic system, $K_h > K_v$, Fig. 4.12, the horizontal dimensions must be multiplied by $\sqrt{\frac{k_v}{k_h}}$ to obtain a transformed section with an isotropic medium. The flownet is drawn and the flow rate is obtained substituting for $K = \sqrt{K_h k_v}$. The flownet is then transferred back to the original anisotropic section by multiplying the horizontal dimensions by $\sqrt{\frac{K_h}{K_v}}$.



a. Flow fields not square



b. orthogonal network
 $Q = \frac{n_f}{n_d} \sqrt{K_h K_v} h$

Source: Raghunath, H. M. (1987). Groundwater. Wiley Eastern.

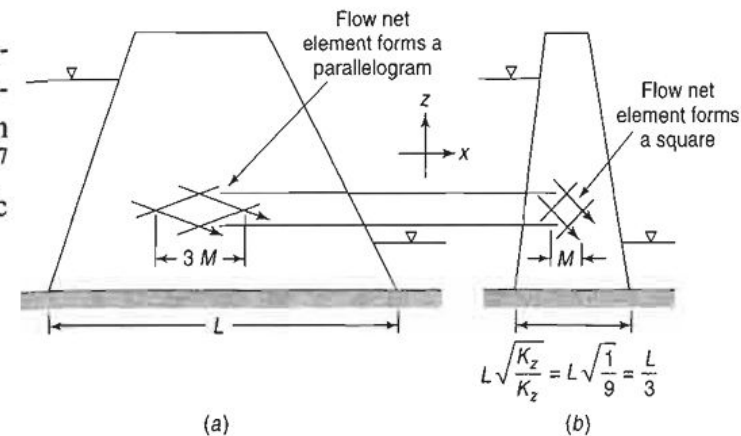


In anisotropic media, flow lines and equipotential lines are not orthogonal except when the flow is parallel to one of the principal directions.⁸ In order to calculate flows for this situation, the boundaries of a flow section must be transformed so that an isotropic medium is obtained. For the typical alluvial case of $K_x > K_z$, all horizontal dimensions are reduced by the ratio $\sqrt{K_z / K_x}$. This creates a transformed section with an isotropic medium having an equivalent hydraulic conductivity

$$K' = \sqrt{K_x K_z} \quad (3.6.7)$$

With this transformed section, the flow net can be drawn and flow rate determined.

After the flow net has been defined, it can be converted back to the true anisotropic section by multiplying all horizontal dimensions by $\sqrt{K_x / K_z}$. Figure 3.6.2 illustrates the procedure for an earth dam as well as demonstrates the distortion created by anisotropy in an element of the flow net. The technique can also be extended to anisotropic two-layer systems.⁶⁷ Figure 3.6.3 shows contrasting flow nets for channel seepage through layered anisotropic media.





Example 4.15: From flownet analysis of a well pumping at 150,000 lpm at Neyveli Lignite Project the following data were obtained:

Number of flow channels	30
Head drop between successive piezometric contours	6 m
Thickness of aquifer	30 m

Determine the transmissibility and permeability of the aquifer.

Solution

$$\begin{aligned} T &= \frac{Q}{n_f \Delta h} = \frac{150,000}{30 \times 6} (60 \times 24) \\ &= 12 \times 10^5 \text{ lpd/m,} \quad \text{or } 1200 \text{ m}^2/\text{day} \\ K &= \frac{T}{b} = \frac{12 \times 10^5}{30} \\ &= 4 \times 10^4 \text{ lpd/m}^2, \quad \text{or } 40 \text{ m/day} \end{aligned}$$

Source: Raghunath, H. M. (1987). Groundwater. Wiley Eastern.



Definition: A flow regime where hydraulic head at any point does not change with time

Assumptions: Inputs (recharge) equal outputs (discharge).

Applicability: Long-term average conditions or analyzing stable pumping scenarios.

Need for Governing Equation

Limitations of Darcy's Law: Darcy's Law ($q = -Ki$) describes flow at a point or 1D column. It does not account for mass balance in 3D heterogeneous systems (Wang & Anderson, 1982).

Modeling: A partial differential equation (PDE) is needed to predict head distributions ($h(x,y,z)$) across an entire aquifer.



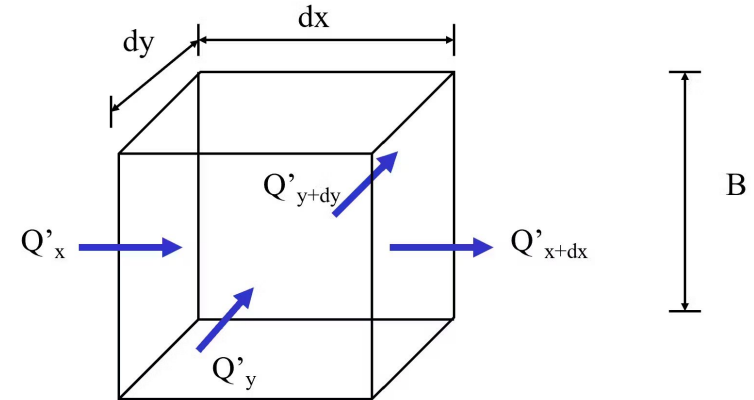
Control Volume Approach: Consider a small cubic element (dx, dy, dz) in the porous medium.

Continuity Equation: Law of Conservation of Mass.

Mass In - Mass Out = Change in Storage

Steady State: Change in Storage = 0, so
Mass In = Mass Out.

Steady state PDE



Control Volume

Fig: Steady flow through a control volume,
Geohydraulics n.d.

Governing Equation of Groundwater Flow: Laplace Equation



From Darcy's law,

$$V = Ki$$

where K = coefficient of permeability of the formation; and i = hydraulic

$$\text{gradient} = \text{grad} \left(z + \frac{P}{w} \right) = \frac{dh}{ds}.$$

Therefore

$$V = -K \frac{dh}{ds} \quad (4.23)$$

the negative sign indicates the ground water flow in the direction of decreasing head, i.e. in the direction s .

It is convenient to introduce a velocity potential ϕ , defined as a scalar function of time and space, such that the velocity components in the x -, y - and z - directions are given by

$$u = -\frac{\partial \phi}{\partial x}, v = -\frac{\partial \phi}{\partial y}, w = -\frac{\partial \phi}{\partial z} \quad (4.24)$$

and generally,

$$V = -\frac{\partial \phi}{\partial s} \quad (4.25)$$

where V is the velocity in the flow direction s .

From Eqs. (4.23) and (4.25),

$$\frac{d\phi}{ds} = K \frac{dh}{ds}$$

or

$$\phi = Kh + C$$

where C = a constant which disappears during differentiation and is

usually taken as zero (since it does not alter the flow pattern in any manner).

Therefore

$$\phi = Kh \quad (4.26)$$

Since ground water can be considered incompressible for all practical purposes, i.e. $\rho = \text{constant}$, the equation of continuity for *steady ground-water flow* is obtained by putting

$$\frac{\partial \rho}{\partial t} = 0 \text{ in Eq. (4.61), as}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (4.27)$$

From Eq. (4.24),

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad (4.28)$$

or,

$$\Delta^2 \phi = 0 \quad (4.29)$$

which is the Laplace equation.

Source: Raghunath, H. M. (1987). Groundwater. Wiley Eastern.

Governing Equation of Groundwater Flow: Laplace Equation



For a homogeneous and isotropic medium (formation), from Eqs. (4.24) and (4.26),

$$u = -K \frac{\partial h}{\partial x}, \quad v = -K \frac{\partial h}{\partial y}, \quad w = -K \frac{\partial h}{\partial z}$$

From Eq. (4.27),

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} + \frac{\partial^2 h}{\partial z^2} = 0 \quad (4.30)$$

$$\text{or} \quad \nabla^2 h = 0 \quad (4.31)$$

i.e. for steady ground water flow in homogeneous isotropic medium, Laplace equation is satisfied, and a flow net can be constructed.

In anisotropic formation, if the permeabilities in the three principal directions are K_x , K_y and K_z , the velocity components are given by Darcy's law as

$$u = -K_x \frac{\partial h}{\partial x}, \quad v = -K_y \frac{\partial h}{\partial y}, \quad w = -K_z \frac{\partial h}{\partial z} \quad (4.32)$$

From the continuity Eq. (4.27), for steady flow,

$$K_x \frac{\partial^2 h}{\partial x^2} + K_y \frac{\partial^2 h}{\partial y^2} + K_z \frac{\partial^2 h}{\partial z^2} = 0 \quad (4.33)$$

If a transformation could be made such that

$$X = \frac{x}{\sqrt{K_x}}, \quad Y = \frac{y}{\sqrt{K_y}}, \quad Z = \frac{z}{\sqrt{K_z}}$$

Eq. (4.33) becomes

$$\frac{\partial^2 h}{\partial X^2} + \frac{\partial^2 h}{\partial Y^2} + \frac{\partial^2 h}{\partial Z^2} = 0 \quad (4.34)$$

and hence Laplace equation is satisfied.

For a two-dimensional anisotropic system $K_x > K_y$, the horizontal dimensions are reduced by multiplying by the factor $\sqrt{K_y/K_x}$, and a flownet is drawn in the transformed section. This creates an isotropic medium having an equivalent $K = \sqrt{K_x K_y}$. The ground water flow is computed by substituting $K = \sqrt{K_x K_y}$ in the discharge equation and then the flownet is transferred back to the original section, which of course will not be an orthogonal network, Fig. 4.12.

Source: Raghunath, H. M. (1987). Groundwater. Wiley Eastern.



Potential Flow Theory assumes that the fluid is **incompressible**, the flow is **irrotational**, and the medium is **homogeneous and isotropic**.

In groundwater hydrology, water flows because of **differences in hydraulic head**.

Potential flow theory assumes that this flow can be described using a **potential function (velocity potential or hydraulic potential)** and the flow follows mathematical laws similar to fluid mechanics and electrostatics.

Darcy's Law and Potential Flow: Darcy's law relates velocity to hydraulic head:

$$\vec{v} = -K\nabla h$$

where:

- $\vec{v}$ = Darcy velocity
- K = hydraulic conductivity
- ∇h = hydraulic gradient

Since velocity is the gradient of a scalar potential (head), groundwater flow is a **potential flow**



For homogeneous, isotropic aquifers under steady-state conditions, the continuity equation (conservation of mass) reduces to Laplace's equation:

$$\nabla^2 h = 0$$

From potential flow theory:

- **Equipotential lines** → lines of constant head
- **Flow lines** → direction of groundwater flow
- They intersect at **right angles (90°)**

This helps in **visualizing groundwater movement**.



Assumptions:

1. **Steady State:** No change in head over time.
2. **Homogeneous:** Hydraulic conductivity (K) is independent of position.
3. **Isotropic:** K is the same in all directions ($K_x = K_y = K_z$).
4. **Incompressible:** Fluid and matrix density are constant.
5. **Conservative:** No internal sources or sinks (recharge/pumping).

Boundary Conditions:

- **Importance:** A PDE cannot be solved without boundary conditions (Wang & Anderson, 1982).
- **Types:** **Dirichlet (First-type):** Constant head (e.g., a lake or river), **Neumann (Second-type):** Constant flux (e.g., an impervious rock layer where flux = 0).

Uses

- **Groundwater Analysis:** Essential for resource quantification and contamination transport.
- **Seepage Problems:** Critical for dam safety (preventing piping/blowout).
- **Design:** Used in designing dewatering systems for excavations and foundations (Das, 2013).
- **Summary:** From streamlines to the Laplace equation, these tools allow us to predict and manage the invisible flow of groundwater.



Assuming flow to be horizontal and uniform everywhere in vertical section ,
discharge per unit width

$$q = V \cdot A = - \frac{d\phi}{ds} (h \cdot 1) \approx -k \left(\frac{dh}{ds} \right) \cdot h \approx -k \left(\frac{dh}{dx} \right) h$$

According to Dupuit ,
In groundwater flow the slope of the water table is small.

That means:

- The flow lines are **almost horizontal**
- Vertical velocity is **negligible**
- Flow direction $\approx$ horizontal direction
- So, $ds \approx dx$

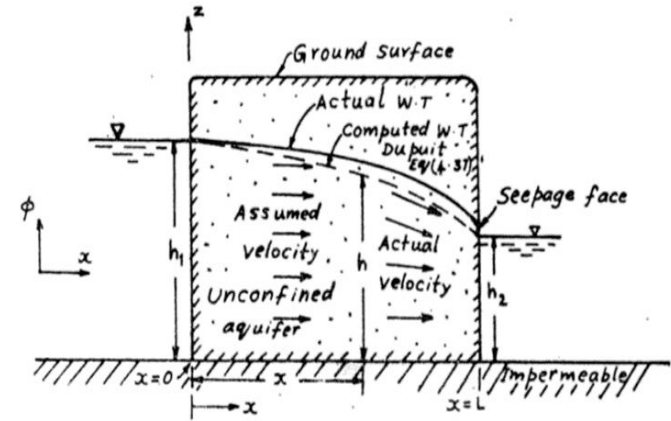


Fig. 4.13 Steady flow in an unconfined aquifer.

Raghunath, H. M. (1987). Groundwater. Wiley Eastern.



The difference between the slanted path and horizontal distance is almost zero.
Mathematically:

$$ds = dx / \cos\theta$$

Since θ is very small, $\cos\theta \approx 1 \Rightarrow ds \approx dx$

In potential flow theory:

ϕ = velocity potential

For groundwater flow,

$\phi \propto h$ (hydraulic head)

So practically:

Change in potential = change in hydraulic head

That's why we often replace ϕ with h in derivations.

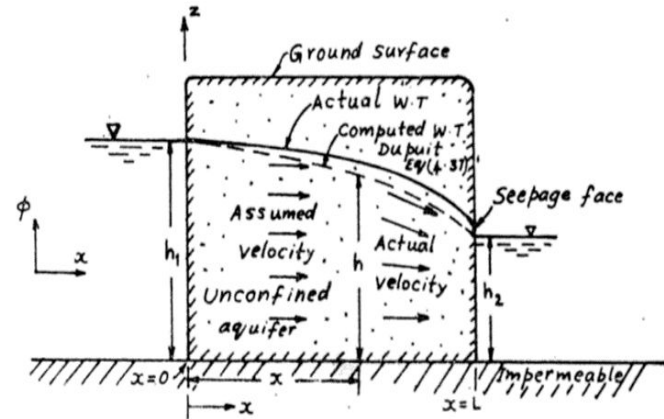


Fig. 4.13 Steady flow in an unconfined aquifer.

Raghunath, H. M. (1987).
Groundwater. Wiley Eastern.



Rearranging (derivation starts)

$$q = -Kh(dh/dx)$$

$$h dh = -(q/K) dx$$

Integrate both sides:

$$\int h dh = -\frac{q}{K} \int dx$$

$$\frac{h^2}{2} = -\frac{q}{K} x + C$$

Multiply both sides by 2:

$$h^2 = -\frac{2q}{K} x + C'$$

When $x=0$, $h=h_1$ and $C = \frac{kh_1^2}{2}$

Therefore,

$$q = \frac{k}{2x} (h_1^2 - h^2)$$

which is the Dupuit's equation and indicates that for a given discharge the water table is parabolic in form.



Applying the boundary condition $h=h_2$ when $x=L$

$$q = \frac{k}{2L} (h_1^2 - h_2^2)$$

The equation for the phreatic line is obtained as

$$h = \sqrt{h_1^2 - (h_1^2 - h_2^2) \frac{x}{L}}$$

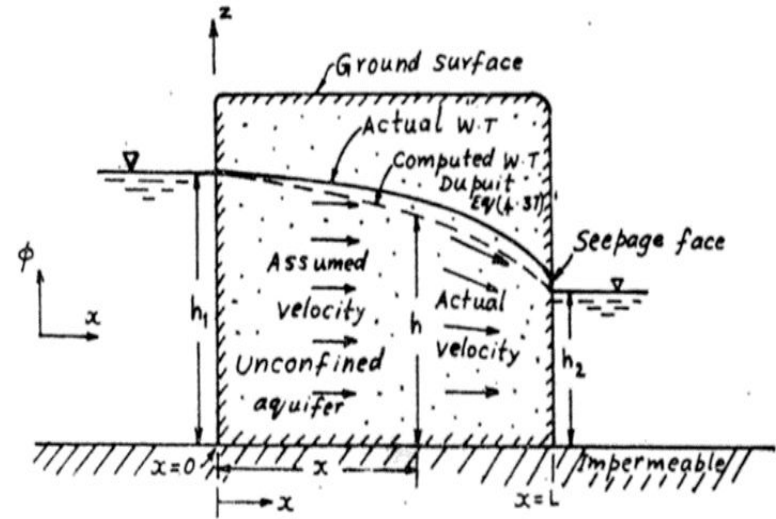


Fig. 4.13 Steady flow in an unconfined aquifer.

Raghunath, H. M. (1987). Groundwater. Wiley Eastern.



- ❖ **The Mismatch:** There is a difference between the water level calculated using the **Dupuit equation** and the **actual water level** measured in the ground.
- ❖ **Location Matters:** This difference (or "departure") is most noticeable in the area **immediately surrounding the well**.
- ❖ **The Higher Water Table:** In reality, the actual water table sits **higher** than what the math predicts.
- ❖ **The Flawed Assumption:** The math assumes water only flows **horizontally**. In real life, water near a well also moves **downward**.



- ❖ **Flow Components:** Because the water has a vertical (downward) speed, it requires a **greater thickness of saturated soil** to move the same amount of water compared to purely horizontal flow.
- ❖ **Slope Errors:** The mathematical assumption that the slope of the water table (dh/ds) is roughly equal to its horizontal change (dh/dx) is **incorrect** near the well.
- ❖ **The Seepage Face:** Because the water table cannot connect directly to the water surface inside the well, it curves and hits the well wall tangentially, forming what is called a **seepage face**.
- ❖ **The "Silver Lining":** Despite these errors in predicting the water level's shape, the Dupuit equation is **still accurate** for calculating the total flow rate (q) or the soil's permeability (K).



For the vertical component of velocity (w),

We have, continuity equation in two dimensions

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0$$

Here,

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left(-K \frac{dh}{dx} \right) = -K \frac{d^2 h}{dx^2}$$

So,
$$w = - \int_0^z \frac{\partial u}{\partial x} dz = -K \int_0^z \frac{\partial^2 h}{\partial x^2} dz = -Kz \frac{d^2 h}{dx^2}$$

We have,

$$q = -Kh \frac{dh}{dx}$$

$$\frac{dh}{dx} = -\frac{q}{Kh}$$



Differentiate w.r.t. x :

$$\frac{d^2 h}{dx^2} = \frac{d}{dx} \left(-\frac{q}{Kh} \right)$$

$$w = -\frac{q^2 z}{Kh^3}$$

Since q and K are constants:

$$\frac{d^2 h}{dx^2} = \frac{q}{Kh^2} \frac{dh}{dx}$$

Substitute $\frac{dh}{dx} = -\frac{q}{Kh}$:

$$\frac{d^2 h}{dx^2} = -\frac{q^2}{K^2 h^3}$$

The vertical velocity component increases with elevation and at the phreatic surface, $z=h$

$$w = -\frac{q^2}{Kh^2}$$



EXAMPLE 4.1.2

A stratum of clean sand and gravel between two channels (see Figure 4.1.2) has a hydraulic conductivity $K = 10^{-1}$ cm/sec, and is supplied with water from a ditch ($h_0 = 6.5$ m deep) that penetrates to the bottom of the stratum. If the water surface in the second channel is 4 m above the bottom of the stratum and its

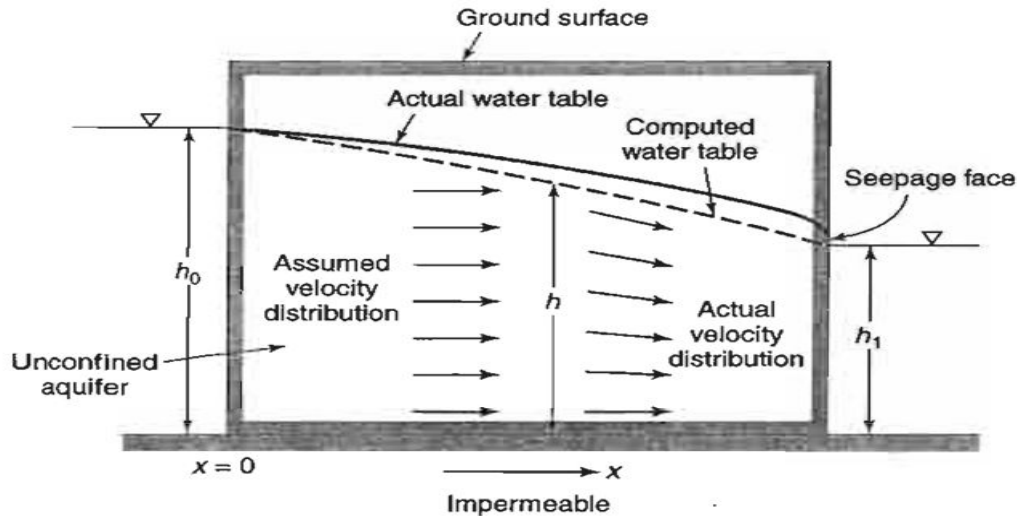


Figure 4.1.2. Steady flow in an unconfined aquifer between two water bodies with vertical boundaries.



distance to the ditch is $x = 150$ m (which is also the thickness of the stratum), estimate the unit flow rate into the gallery.

SOLUTION

The flow is computed using the Dupuit equation (4.1.6) for unit flow, where

$$K = 10^{-1} \text{ cm/sec} = 86.4 \text{ m/day}$$

$$q = \frac{K}{2x} (h_0^2 - h^2) = \frac{86.4 \text{ m/day}}{2(150 \text{ m})} (6.5^2 - 4^2) \text{ m}^2 = 7.56 \text{ m}^2/\text{day}$$

Todd, D. K., & Mays, L. W. (2005). Groundwater hydrology (3rd ed.). John Wiley & Sons.

Depth of GWT above impermeable soil layer = H

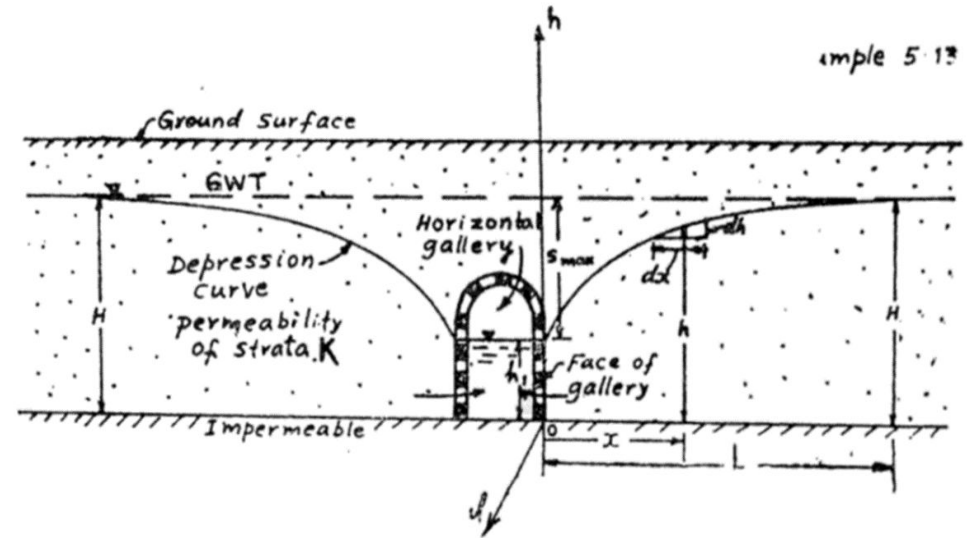
Depth of water table in the gallery = h_1

The phreatic surface is parabolic dropping from H to h in a distance L .

The quantity of water flowing into the trench from one side per unit length of shoreline is

$$q = VA = Ki(h \cdot 1) = K(dh/dx) \cdot h$$

x being measured from the face of the gallery.



Raghunath, H. M. (1987). Groundwater. Wiley Eastern.



Flow into Horizontal Galleries extending up to Impervious Soil Layer

Rearranging, we get,

$$q \cdot dx = k \cdot h \cdot dh$$

Integrating

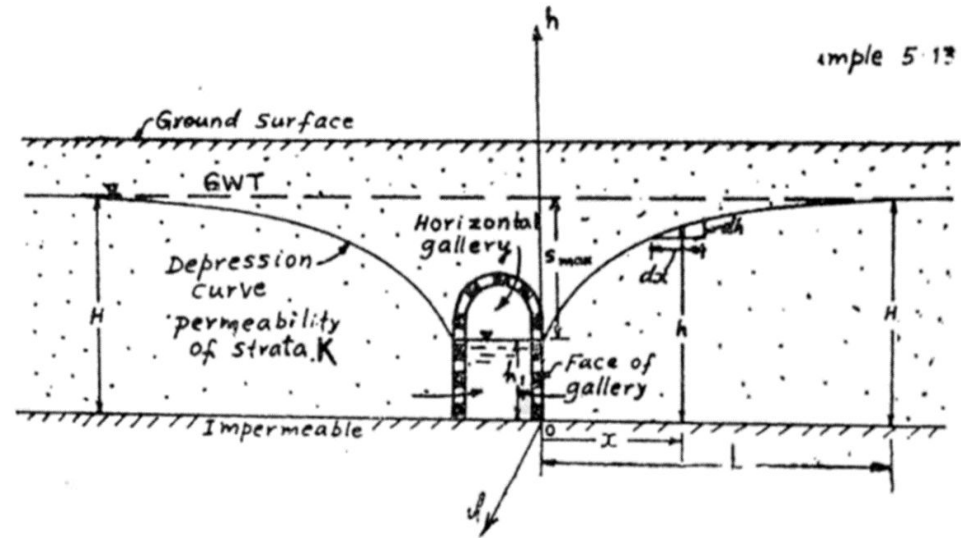
$$qx = \frac{kh^2}{2} + c_1$$

when $x = 0$, $h = h_1$; $c_1 = -\frac{Kh_1^2}{2}$

$$q = \frac{K}{2x} (h^2 - h_1^2)$$

which is parabolic. Putting $h = H$ when $x = L$

$$q = \frac{K}{2L} (H^2 - h_1^2)$$



Raghunath, H. M. (1987). Groundwater. Wiley Eastern



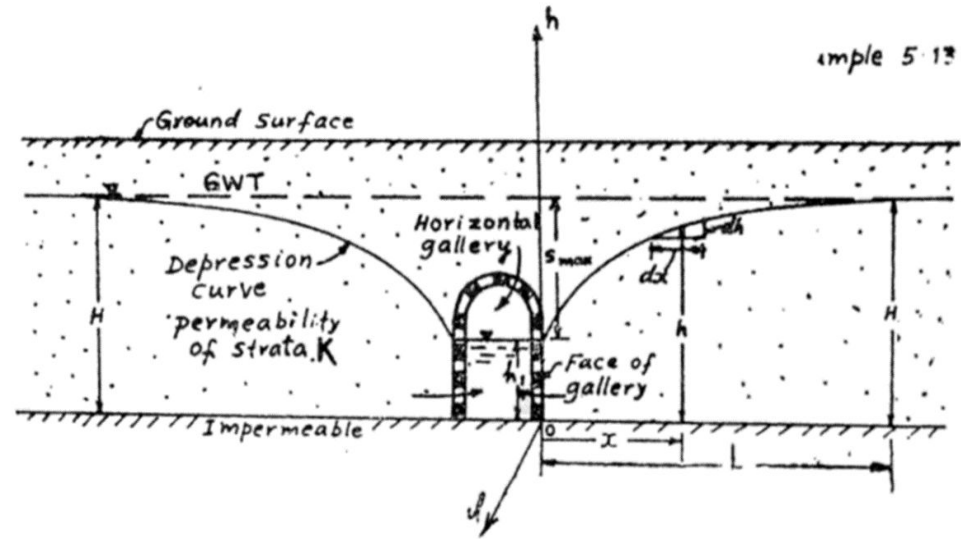
Flow into Horizontal Galleries extending up to Impervious Soil Layer

From above two equations,

$$h = \sqrt{h_1^2 + \frac{x}{L} (H^2 - h_1^2)}$$

The quantity of water flowing into the gallery of length l from both sides is given as:

$$Q = 2ql = \frac{Kl}{L} (H^2 - h_1^2)$$



Raghunath, H. M. (1987). Groundwater. Wiley Eastern.

When the watertable in the gallery drops, i.e. h_1 decreases, L increases and upon increasing h_1 , L decreases.



Flow into Horizontal Galleries extending up to Impervious Soil Layer

As an example, given $H = 7$ m, $h_1 = 2$ m, $L = 400$ m, $K = 60$ m/day, $l = 200$ m.

Flow from one side

$$q = \frac{K}{2L} (H^2 - h_1^2) = \frac{60}{2 \times 400} (7^2 - 2^2) = 3.375 \text{ m}^3/\text{day per m}$$

(i) Flow into the gallery $Q = 2ql = 2 \times 3.375 \times 200$
 $= 1350 \text{ m}^3/\text{day}$

(ii) Equation to the phreatic surface

$$h = \sqrt{h_1^2 + \frac{x}{L} (H^2 - h_1^2)}$$

$$h = \sqrt{2^2 + \frac{x}{400} (7^2 - 2^2)}$$

or,

$$h = \sqrt{4 + 0.1125x}$$

The GWT at 100 m from the face of the gallery is

$$h = \sqrt{4 + 0.1125 \times 100} = 3.78 \text{ m}$$

or drawdown $s = H - h = 7 - 3.78 = 3.22$ m.

(iii) The maximum drawdown at the gallery is

$$s_{\max} = H - h_1 = 7 - 2 = 5 \text{ m}$$

Raghunath, H. M.
 (1987). Groundwater.
 Wiley Eastern.

- An unconfined aquifer has a free water table under **atmospheric pressure** (*Raghunath, 2006*).
- Uniform recharge due to **rainfall** or **infiltration** adds water to the aquifer.
- Under steady-state conditions, hydraulic head varies **spatially** but not with time.
- Flow area varies with saturated thickness, affecting velocity distribution (*Bear, 1979*).
- The water table assumes a **parabolic profile** under uniform recharge.

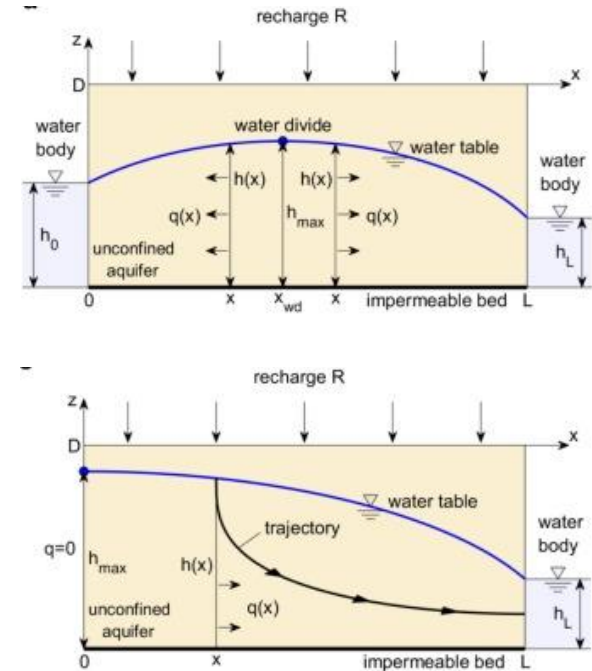


Fig: Parabolic Profile under Recharge

Source: <https://www.sciencedirect.com>

We know,

$$dq = P \cdot dx \dots\dots(i)$$

$$q = -kh \frac{dh}{dx} = -\frac{1}{2}k \frac{d(h^2)}{dx}$$

$$\frac{dq}{dx} = -\frac{1}{2}k \frac{d^2(h^2)}{dx^2} = P, \text{ from (i)}$$

Therefore,

$$\frac{d^2(h^2)}{dx^2} = -\frac{2P}{k} \dots\dots(1)$$

Further,

$$\frac{dq}{dx} = -k \frac{d}{dx} \left(h \frac{dh}{dx} \right) = P \dots\dots(2)$$

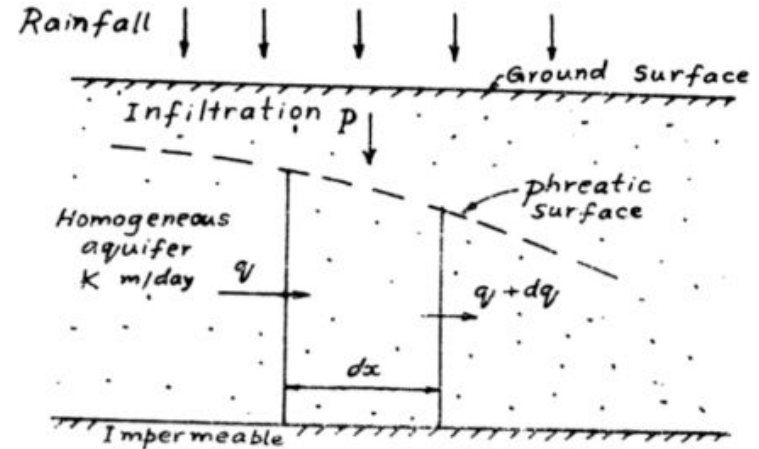


Fig. 4.15 Flow in an unconfined aquifer with rainfall.

Source: Raghunath, 2006

Unconfined Aquifer with Recharge



The solution is: $kh^2 + Px^2 = c_1x + c_2$

From the boundary conditions:

at $x = 0, h = h_1$ and at $x = L, h = h_2$;
the equation for phreatic surface is:

$$h = \sqrt{h_1^2 - \frac{(h_1^2 - h_2^2)}{L} x + \frac{P}{k} (L - x) x} \dots\dots(3)$$

And,

$$q_x = q_0 + Px \dots\dots(4)$$

where, q_0 is at $x = 0$.

Integration of eqn (3) yields

$$q_0 = k \frac{(h_1^2 - h_2^2)}{2L} - \frac{PL}{2}$$

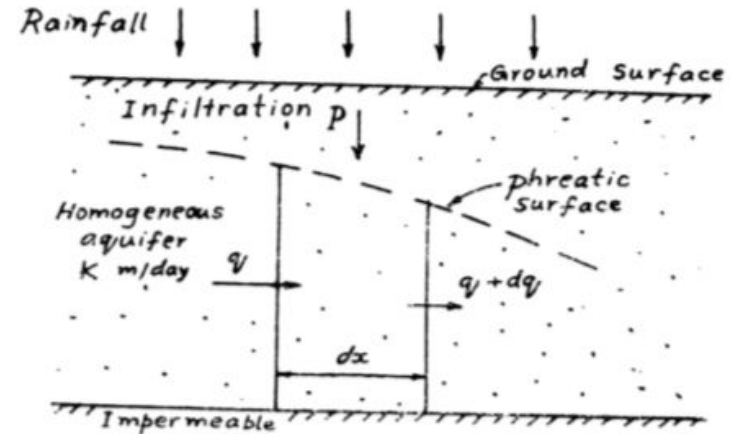


Fig. 4.15 Flow in an unconfined aquifer with rainfall.

Source: Raghunath, 2006

From eqn (4),

$$q_x = k \frac{(h_1^2 - h_2^2)}{2L} - P \left(\frac{L}{2} - x \right) \dots\dots(5)$$

If there is no recharge by rain, then

$$\frac{dq}{dx} = 0, \quad \frac{d^2(h^2)}{dx^2} = 0$$

where, q and h are determined as:

$$q = \frac{k}{2L} (h_1^2 - h_2^2)$$

$$h = \sqrt{h_1^2 - \frac{(h_1^2 - h_2^2)}{L} x}$$

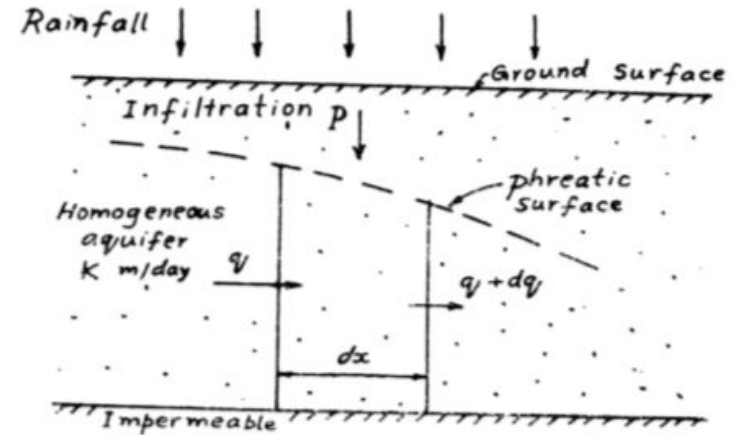


Fig. 4.15 Flow in an unconfined aquifer with rainfall.

Source: Raghunath, 2006

Example 4.16: Rainfall at the rate 10 mm/hr falls on a strip of land 1 km wide laying between two parallel canals with 2 m difference in their water surface levels, Fig. 4.16. It is underlain by a horizontal impermeable stratum at 10 m below the water surface of the lower canal. Assuming a permeability of 12 m/day with vertical boundaries and all the rainfall infiltrates into the soil, compute the discharge per metre length into both the canals.

Solution

$$q = AKi = -(h \cdot 1) K \frac{dh}{dx} = -\frac{1}{2} K \frac{d(h^2)}{dx}$$

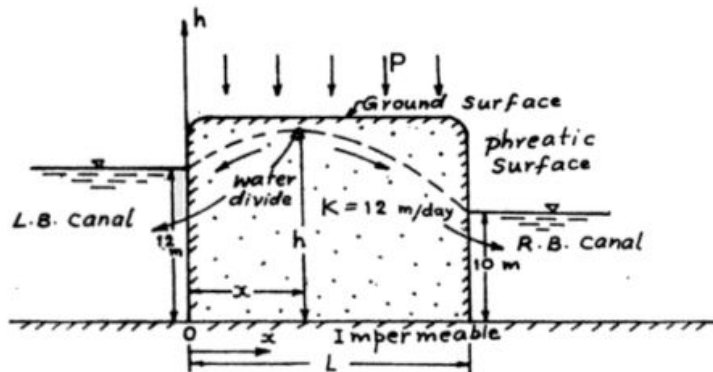


Fig. 4.16 Aquifer between parallel canals recharged by rainfall, Example 4.16.

From Eq. (4.43), $\frac{d^2(h^2)}{dx^2} = -\frac{2P}{K}$

$$\frac{d(h^2)}{dx} = -\frac{2P}{K} x + c_1$$

$$h^2 = -\frac{P}{K} x^2 + c_1 x + c_2$$

Taking a reference origin as 0, Fig. 4.16,

when $x = 0, h = 12$: $12^2 = -0 + 0 + c_2, \quad c_2 = 144$

when $x = 1000, h = 10$: $10^2 = -\frac{0.01 \times 24}{12} \times 10^6 + 1000 c_1 + 144$

$$c_1 = 20$$

$$q = -\frac{1}{2} K \frac{d(h^2)}{dx} = -\frac{1}{2} K \left(-\frac{2P}{K} x + c_1 \right)$$

At $x = 0, \quad q_{L.B. \text{ Canal}} = -\frac{12}{2} (-0 + 20) = -120 \text{ m}^3/\text{day/m}$

the negative sign indicates flow towards left.

At $x = 1000,$

$$q_{R.B. \text{ Canal}} = -\frac{12}{2} \left(-\frac{2(0.01 \times 24)}{12} \times 1000 + 20 \right) = 120 \text{ m}^3/\text{day/m}$$



So, there is a discharge of $120 \text{ m}^3/\text{day}$ into both the canals, for each metre length of the strip of land.

Note (i): Recharge into land = Discharge into both the canals

$$= (0.01 \times 24)(1000 \times 1) = 120 + 120 = 240 \text{ m}^3/\text{d/m}$$

$$= 240 \text{ m}^3/\text{d/m}$$

Note (ii): If there were no recharge by rain, seepage from the upper canal to the lower canal, Eq. (4.36),

$$q = \frac{K}{2L} (h_1^2 - h_2^2) = \frac{12}{2 \times 1000} (12^2 - 10^2) = 0.264 \text{ m}^3/\text{day/m}$$

Note (iii): With recharge, the equation to phreatic surface, Eq. (4.45),

$$h = \sqrt{144 + 19.956x - 0.02x^2}$$

and the water table divide is obtained by putting $dh/dx = 0$,

$$x = 498.9 \text{ m}$$

where

$$h = 71.6 \text{ m}$$



- A confined aquifer is bounded by impervious layers and fully saturated.
- Aquifer thickness remains constant along the flow direction.
- Groundwater flows due to **piezometric head difference** (*Raghunath, 2006*).
- Flow area remains constant, producing a uniform hydraulic gradient.
- The piezometric surface is a **straight line** under steady flow (*Bear, 1979*).

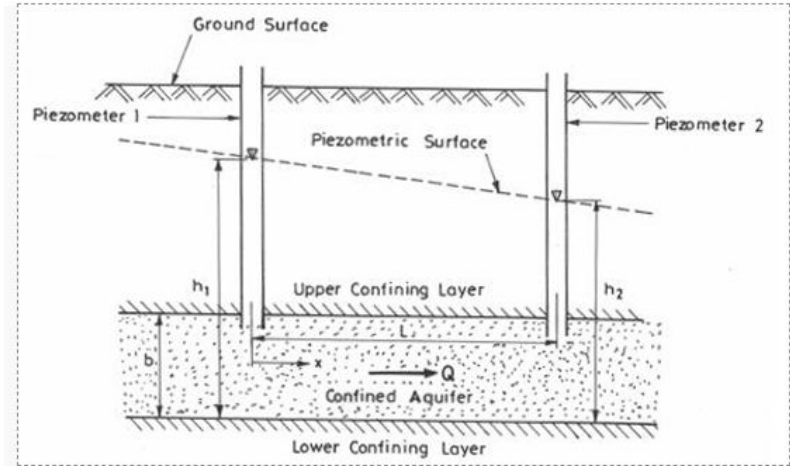


Fig. 1.1. Steady flow through a confined aquifer of uniform thickness.

(Modified from Fetter, 1994)

Source: <https://dspmuranchi.ac.in>

We have,

Discharge per unit width:

$$q = (b \cdot 1)v = b \left(-\frac{d\phi}{ds} \right) = -b \cdot k \frac{dh}{dx}$$

For Steady Flow,

$$\frac{dq}{dx} = -bk \frac{d^2h}{dx^2}$$

Since, b and k have finite values,

$$\frac{d^2h}{dx^2} = 0$$

which has the solution:

$$h = c_1x + c_2$$

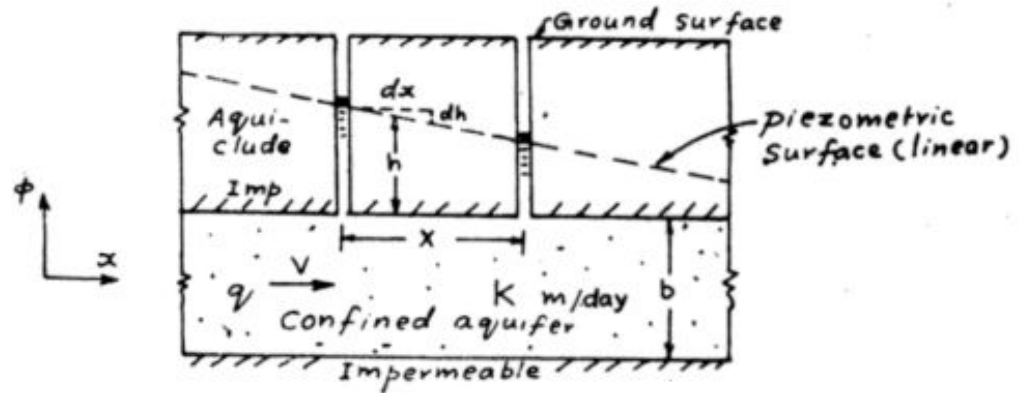


Fig. 4.17 Unidirectional flow in a confined aquifer of constant thickness.

Source: Raghunath, 2006



Selecting the datum such that $h = 0$ when $x = 0$, and also from

$$\frac{dh}{dx} = -\frac{q}{bk} = -\frac{v}{k}$$

$$\text{or, } h = -\frac{v}{k}x \dots\dots(6)$$

which indicates that the head decreases linearly with x , i.e. the direction of the flow.

Knowing the piezometric levels h_1 and h_2 , X apart, the discharge per unit width of the aquifer is given by:

$$q = b.k \frac{h_1 - h_2}{x} \dots\dots(7)$$

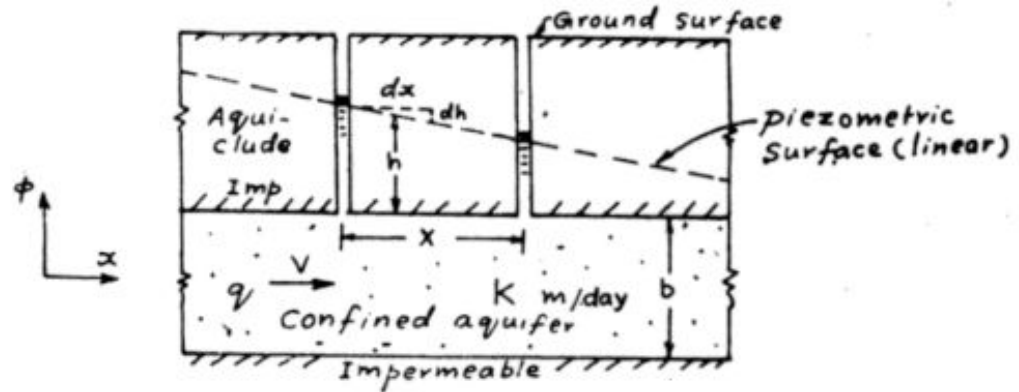


Fig. 4.17 Unidirectional flow in a confined aquifer of constant thickness.

Source: Raghunath, 2006



- Confined aquifer thickness varies due to **geological irregularities**.
- Flow remains steady and essentially horizontal (*Bear, 1979*).
- Variation in thickness alters **flow area** and **hydraulic gradient**.
- Groundwater velocity changes spatially along the aquifer.
- The piezometric surface becomes **curved** under steady flow conditions.

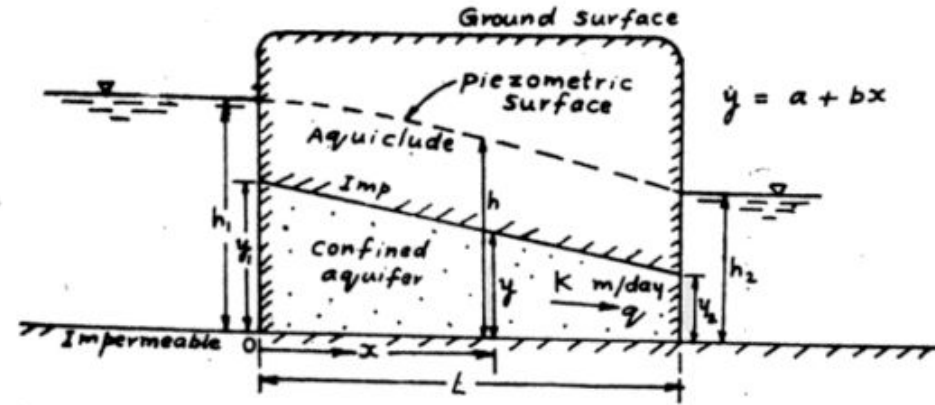


Fig. 4.18 Flow through confined aquifer of variable thickness.

Source: Raghunath, 2006

We have,

Discharge per unit width:

$$q = v (y.1) = -\frac{\partial \phi}{\partial s} y = -k \frac{dh}{dx} y$$

$$dh = -\frac{q}{ky} dx, \quad y = a + bx \text{ (say),}$$

$$\text{So, } h = -\frac{q}{kb} \ln (a + bx) + c \dots\dots(8)$$

where, q and c can be determined by the boundary conditions.

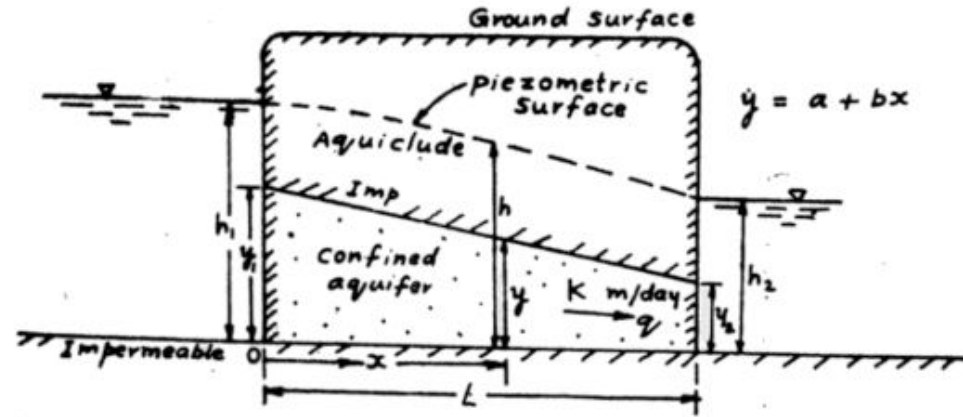


Fig. 4.18 Flow through confined aquifer of variable thickness.

Source: Raghunath, 2006



For example given $h_1 = 10$ m, $h_2 = 7$ m, $y_1 = 6$ m, $y_2 = 4$ m, $L = 400$ m,
 $K = 3 \times 10^{-6}$ m/s,

$$y = 6 - \frac{(6-4)x}{400} = 6 - 0.005x$$

$$h = - \frac{q}{(3 \times 10^{-6})(-0.005)} \ln(6 - 0.005x) + c$$

$$x = 0, h = 10 = \frac{q}{1.5 \times 10^{-8}} \ln 6 + c$$

$$x = 400, h = 7 = \frac{q}{1.5 \times 10^{-8}} \ln 4 + c$$

$$\text{subtracting, } 3 = \frac{q}{1.5 \times 10^{-8}} \ln \frac{6}{4}$$

$$\text{or } q = 1.11 \times 10^{-7} \text{ m}^3/\text{s per width, or metre m}^2/\text{s}.$$

Alternatively, by taking average values

$$q = KiA = (3 \times 10^{-6}) \frac{10-7}{400} \left(\frac{(6+4)}{2} \times 1 \right) = 1.125 \times 10^{-7} \text{ m}^2/\text{s}$$

which is only 1.35% higher than the previous result.



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THANK YOU !!!